Rational Numbers and Order of Operations

Concept

Rational numbers can be written as fractions of integers with non-zero denominators.

Decimals

Terminating and repeating decimals are rational.

Signs

Subtracting means adding the opposite; sign rules differ for products.

Powers

$(-3)^2 = 9$, but $-3^2 = -9$.

Order

Brackets → powers → multiplication/division → addition/subtraction.

Example 1

Evaluate $-3/4 \div 1/2 + 2/3$.

Solution

1. Divide first: $(-3/4)(2) = -3/2$.

2. Add: $-9/6 + 4/6 = -5/6$.

Rational Numbers and Order of Operations

Example 2

Evaluate $1 - 2(3/5)^2$.

Solution

1. Square: $(3/5)^2 = 9/25$.

2. Multiply and subtract: $1 - 18/25 = 7/25$.

Rational Numbers and Order of Operations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Order $-2/3$, -0.7 and $-3/5$ from least to greatest.

2. Evaluate $-5/6 + 3/4$.

3. Evaluate $(2 - 5/2)^2$.

4. Evaluate $-4 + 12 \div (-3)$.

5. Explain why $0.272727\ldots$ is rational.

6. Evaluate $[3/4 - 1/2] \div (-1/8)$.

Rational Numbers and Order of Operations

Practice answers | check after trying

1. $-0.7 < -2/3 < -3/5$, using $-0.700\ldots$, $-0.666\ldots$ and $-0.600\ldots$.

2. $-10/12 + 9/12 = -1/12$.

3. $(-1/2)^2 = 1/4$.

4. $-4 - 4 = -8$.

5. Let $x = 0.272727\ldots$. Then $100x - x = 27$, so $x = 27/99 = 3/11$.

6. $(1/4) \div (-1/8) = -2$. Check: $(-2)(-1/8) = 1/4$.

Whole-Number Exponent Laws

Concept

Exponent laws simplify repeated factors. They do not combine unlike terms in sums.

Product / quotient

$a^m a^n = a^{m+n}$; $a^m / a^n = a^{m-n}$ for $a \neq 0$.

Power of a power

$(a^m)^n = a^{mn}$.

Product power

$(ab)^n = a^n b^n$; $a^0 = 1$ for $a \neq 0$.

Warning

$(a + b)^2$ is not $a^2 + b^2$. Grade 9 uses whole-number outcomes.

Example 1

Simplify $(2a^3b)^2$.

Solution

1. Square every factor: $2^2 (a^3)^2 b^2$.
2. Result: $4a^6b^2$.

Whole-Number Exponent Laws

Example 2

Simplify $(18x^5y^3)/(6x^2y)$.

Solution

1. State restrictions: $x \neq 0$ and $y \neq 0$.

2. Divide coefficients and subtract exponents: $3x^3y^2$.

Whole-Number Exponent Laws

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify m^3m^5.

2. Simplify $(3p^2)^3$.

3. Simplify $20a^6/(5a^2)$, stating restrictions.

4. Evaluate $(-3)^2 + 2^0$.

5. Is $(x + 2)^2 = x^2 + 4$? Disprove with a value.

Whole-Number Exponent Laws

Practice answers | check after trying

1. m^8.

2. $27p^6$.

3. $4a^4$, $a \neq 0$.

4. $9 + 1 = 10$.

5. No: $x = 1$ gives 9 on the left and 5 on the right. The missing term is $4x$.

Polynomial Operations

Concept

Polynomials combine terms with non-negative integer variable exponents.

Degree

The highest variable exponent after simplification.

Like terms

Same variable part; add or subtract their coefficients.

Subtract

Distribute the minus sign to every term in the bracket.

Distribute

Multiply or divide every term by the monomial; keep restrictions.

Example 1

Simplify $(5x - 2) - (3x + 8)$.

Solution

1. Remove brackets: $5x - 2 - 3x - 8$.

2. Collect like terms: $2x - 10$.

Polynomial Operations

Example 2

Expand $-2x(3x - 4)$.

Solution

1. Multiply each term: $(-2x)(3x) + (-2x)(-4)$.
2. Result: $-6x^2 + 8x$.

Example 3

Simplify $(12x^2 - 18x)/(6x)$.

Solution

1. $x \neq 0$. Divide each term by $6x$.
2. Result: $2x - 3$, with $x \neq 0$ retained.

Polynomial Operations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. State degree and number of terms of $7 - 2x^2 + x$.

2. Simplify $(4x^2 - x + 2) - (x^2 + 3x - 5)$.

3. Expand $3x(2x + 5)$.

4. Simplify $(15x^2 + 10x)/(5x)$.

5. Explain why $2x + 3x^2$ is not $5x^3$.

Polynomial Operations

Practice answers | check after trying

1. Degree 2; three terms, so a trinomial.

2. $3x^2 - 4x + 7$.

3. $6x^2 + 15x$.

4. $3x + 2$, $x \neq 0$.

5. The terms are unlike and are being added, not multiplied. At $x = 2$ the original is 16 while $5x^3$ is 40.

Multi-Step Linear Equations

Concept

A multi-step linear equation is solved by preserving equality at each step.

Plan

Expand → collect terms → isolate the variable → check.

Fractions

Multiply every term on both sides by the LCD.

No solution

A contradiction such as $0 = 5$.

All values

An identity such as $0 = 0$, on the allowed domain.

Example 1

Solve $3(x - 2) = 2x + 5$.

Solution

1. Expand: $3x - 6 = 2x + 5$.
2. Subtract $2x$ and add 6: $x = 11$.
3. Check: $3(11 - 2) = 27 = 2(11) + 5$.

Multi-Step Linear Equations

Example 2

Solve $x/3 - 1/2 = x/6 + 1$.

Solution

1. Multiply every term by 6: $2x - 3 = x + 6$.
2. Collect terms: $x = 9$.
3. Check: $3 - 0.5 = 1.5 + 1 = 2.5$.

Multi-Step Linear Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $4(2x - 1) = 3x + 16$.

2. Solve $(x + 1)/4 = (x - 2)/3$.

3. Solve $5x - 7 = 5x + 2$.

4. Solve $3(x + 2) = 3x + 6$.

5. Two consecutive integers total 47. Find them and verify.

Multi-Step Linear Equations

Practice answers | check after trying

1. $8x - 4 = 3x + 16$ gives $5x = 20$, so $x = 4$.

2. $3x + 3 = 4x - 8$ gives $x = 11$.

3. No solution: $-7 = 2$ is false.

4. All real x: both sides are identical.

5. $n + (n + 1) = 47$ gives $n = 23$. The integers are 23 and 24, and their sum is 47.

Linear Relations and Predictions

Concept

A linear relation has constant change in y for equal changes in x.

Rule

$y = mx + b$; m = rate of change, b = output when $x = 0$.

Graph

Label axes and units; connect points only for continuous situations.

Prediction

Interpolation is within observed inputs; extrapolation is beyond them.

Special lines

$y = k$ is horizontal; $x = k$ is vertical.

Example 1

A tank follows $V = 30 - 2t$ litres. When is it empty?

Solution

1. Empty means $V = 0$: $30 - 2t = 0$.

2. $t = 15$ minutes. Meaningful domain: $0 \le t \le 15$.

Linear Relations and Predictions

Example 2

Find a line through (2,7) and (6,15), then estimate y at $x = 4$.

Solution

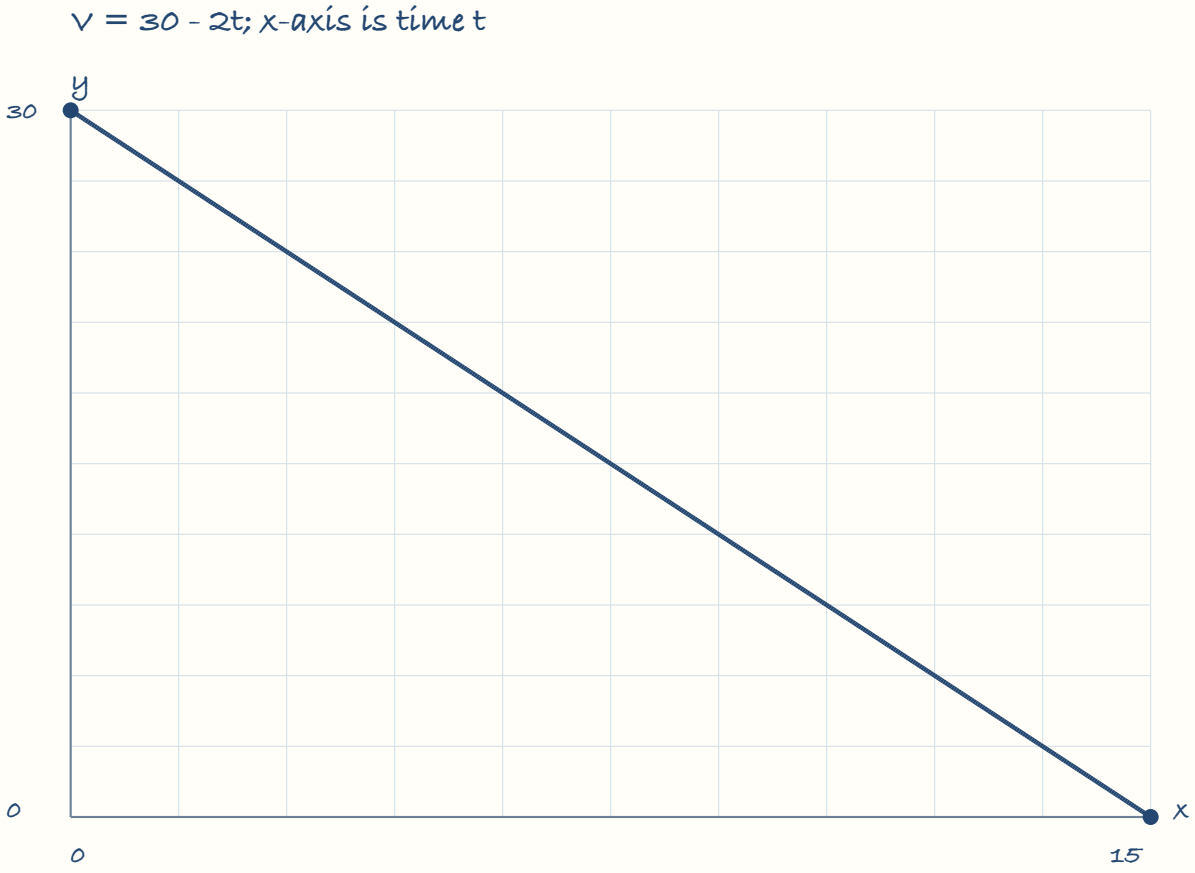
1. Rate $= (15 - 7)/(6 - 2) = 2$.

2. $7 = 2(2) + b \rightarrow b = 3$. Rule: $y = 2x + 3$.

3. At $x = 4$, $y = 11$. This is interpolation.

Linear Relations and Predictions

Visual notes | read the graph



The useful segment ends at 15 minutes, when the tank is empty. Axis endpoints give the scale.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Linear Relations and Predictions

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Make a three-point table for $y = x/2 - 1$ using $x = 0, 2, 4$.

2. Is $(3, 8)$ on $y = 2x + 2$?

3. Describe $x = -4$ and $y = 5$.

4. Observations cover ages 10–15. Is a prediction at age 12 interpolation or extrapolation? What about age 30?

Linear Relations and Predictions

Practice answers | check after trying

1. Points (0,-1), (2,0), (4,1).

2. Yes: 2(3) + 2 = 8.

3. A vertical line through $x = -4$ and a horizontal line at $y = 5$.

4. Age 12 is interpolation; age 30 is extrapolation and may be unreliable because growth patterns can change.

Scale and Similarity

Concept

Similar shapes have equal matching angles and proportional matching lengths.

Scale factor

k = image length/original length.

Scale drawing

1:200 means 1 unit on paper represents 200 matching units.

Match sides

Use corresponding sides in the same order in every ratio.

Metric units

1 m = 100 cm; convert before comparing lengths.

Example 1

Sides 6, 8, 10 are enlarged so the side 6 becomes 9.

Solution

1. Scale factor = 9/6 = 1.5.

2. Other sides: 8(1.5) = 12 and 10(1.5) = 15.

Scale and Similarity

Example 2

A 1.5 m stick casts a 2 m shadow. A tree casts a 12 m shadow.

Solution

1. Assume level ground and the same sun angle: $h/12 = 1.5/2$.

2. $h = 12(1.5/2) = 9$ m.

Scale and Similarity

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Enlarge sides 4, 7, 9 by scale factor 2.5.

2. On a 1:50 plan, what actual distance does 8 cm represent?

3. Corresponding sides are 12 and 18. If another original side is 10, find the corresponding enlarged side.

4. Two rectangles are 4 by 6 and 6 by 8. Are they similar?

Scale and Similarity

Practice answers | check after trying

1. 10, 17.5 and 22.5.

2. $8 \times 50 = 400$ cm $= 4$ m.

3. $k = 18/12 = 1.5$, so the new side is 15.

4. No: $6/4 = 1.5$ but $8/6 = 4/3$. The corresponding scale factors differ, even though all angles are right angles.

Statistics in Society

Concept

Statistics describe data. Good conclusions depend on how the data were collected.

Population / sample

Whole group of interest / subset actually studied.

Sampling

Random and stratified methods can reduce selection bias.

Bias

Check wording, non-response, timing, missing groups and graph scales.

Ethics

Protect privacy; correlation alone does not establish causation.

Example 1

A school surveys only students at the bus stop about transit use. Evaluate it.

Solution

1. Problem: convenience sample overrepresents transit users.
2. Improve: randomly select students from the whole school or within grades.

Statistics in Society

Example 2

Of 200 invited students, 120 respond and 72 prefer A. Interpret the result.

Solution

1. Among respondents: 72/120 = 60% prefer A.
2. The other 80 did not respond; whole-school support remains uncertain.
3. Do not present 60% as a guaranteed population result.

Statistics in Society

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Identify the population and sample in a survey of 60 randomly selected Grade 9 students from one school.

2. Rewrite: 'Don't you agree our amazing new timetable is better?' neutrally.

3. Does an association between ice-cream sales and swimming accidents prove ice cream causes accidents?

4. Give two checks before sharing a chart online.

Statistics in Society

Practice answers | check after trying

1. Population: all Grade 9 students at that school; sample: the selected 60.

2. 'Compared with the previous timetable, how do you rate the new timetable?' Include balanced options and 'no opinion'.

3. No. Hot weather can increase both swimming and ice-cream purchases; association does not establish causation.

4. Check the source and date, axis scale, units, sample method and size, missing groups, and whether the conclusion fits the data. Any two with an explanation are suitable.

Budgets, Banking and Simple Interest

Concept

A budget matches income, expenses and savings over the same time period.

Balance

Income − expenses = surplus or deficit.

Transactions

Add deposits; subtract withdrawals and fees.

Simple interest

$I = Prt$; $A = P + I$. Annual r needs t in years.

Savings goal

(Goal − current savings)/number of periods.

Example 1

Find simple interest on $900 at 4% annually for 8 months.

Solution

1. $t = 8/12$ year; $r = 0.04$.

2. $I = 900(0.04)(8/12) = \$24$; $A = \$924$.

Budgets, Banking and Simple Interest

Example 2

You need $480 in 6 months and have $150. How much must you save monthly?

Solution

1. Amount still needed = 480 − 150 = $330.

2. Monthly saving = 330/6 = $55, ignoring interest.

Budgets, Banking and Simple Interest

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the surplus if income is $850 and expenses total $735.

2. Find simple interest on $1200 at 3.5% annually for 9 months.

3. Convert a $15 weekly expense to an average monthly expense.

4. A $720 goal is 8 months away and $200 is saved. Find monthly saving needed.

Budgets, Banking and Simple Interest

Practice answers | check after trying

1. $115.

2. 1200(0.035)(9/12) = $31.50; final amount $1231.50.

3. 15 × 52/12 = $65 per average month.

4. (720 − 200)/8 = $65 per month. If the budget cannot support this, adjust the timeline, goal or other expenses.